



The LLM Interface Economy: How GPTs Will Replace Browsers and Reshape Commerce

Executive Summary

The way we access information and transact online is undergoing a once-in-a-generation shift. Over the past 25 years, digital commerce has centered on the web browser: users search for products or answers, click through websites, and complete purchases via online checkouts. This white paper argues that this browser-to-search-to-website paradigm is nearing extinction. In its place rises a new **LLM Interface Economy** – a paradigm where large language models (LLMs) and AI **agents** (such as GPT-4, ChatGPT, Google's Gemini, Microsoft Copilots, etc.) become the primary interface for digital activities and transactions.

Historical transitions in the web's evolution offer context for this shift. Just as the open web supplanted walled gardens like AOL in the 1990s, and as search engines and mobile apps redefined user behavior in the 2000s, AI agents are poised to become the next dominant interface. We are entering what some have called *Web 4.0* or the **Agentic Web**, in which software agents act on our behalf. In this new era, users will increasingly **"summon" information and services through natural-language prompts** rather than manually navigating websites ¹ ². The emergence of LLM-driven assistants is as transformative today as the introduction of the domain name system was in 1994 – it creates a new transactional layer and gateway that every business must learn to leverage or risk obscurity.

Key findings and arguments of this thesis include:

- **From Web 1.0 to the Agentic Web:** We outline the milestones of Web 1.0 (static sites and domain names), Web 2.0 (social media, mobile apps), and Web 3.0 (the semantic and decentralized web), and show how LLM-based agents represent the next paradigm shift. This *Web 4.0* is defined by AI agents that can interpret user intent and execute complex tasks across the internet. Microsoft's chief communications officer frames this vision as an **"open agentic web, where AI agents make decisions and perform tasks on behalf of users"** ³.
- **LLMs as the New User Interface:** In the LLM economy, the chat prompt replaces the search box and the browser window. Users will converse with AI assistants that orchestrate information and services from many sources. The **browser itself may recede into the background, becoming a "museum" for static content**, while dynamic "context engines" assemble answers and actions on the fly ⁴. Instead of browsing a dozen websites to plan a trip or buy a product, users will rely on an agent to gather options, answer questions, and complete transactions – all within a single conversational thread. Early evidence of this shift is visible in the rapid uptake of conversational AI: by late 2024, ChatGPT was logging around **1 billion interactions per day** and had surpassed Bing in total traffic ⁵ ⁶. Surveys in 2025 show **over half of U.S. adults have used an LLM tool, and two-thirds of those users treat them "like search engines"** for information retrieval ⁷.

- **Death of the Traditional Search-&Click Workflow:** As AI answers become ubiquitous, the value of search engine result pages and clicks is declining. **Bain & Company finds that 80% of consumers now rely on “zero-click” AI summaries in at least 40% of searches, reducing organic web traffic by an estimated 15–25%** ⁸. In parallel, platforms like ChatGPT and Google’s Bard are encroaching on queries that formerly led to websites – by one estimate, **LLM-based search could overtake traditional search in volume by 2028–2030**, marking a historic inflection point ⁹ ¹⁰. Google’s own leaders acknowledge this trajectory; the head of Google Search has noted that the classic search page will become *“less prominent over time” as AI takes center stage* ¹¹. This means fewer opportunities for businesses to acquire customers via SEO and paid search ads, and a need to adapt content strategies for *AI-driven discovery* rather than simple keyword ranking.
- **Winners and Losers in an Agentic Commerce Era:** We present case studies of companies positioning themselves to thrive in this new paradigm versus those at risk of falling behind. On the winning side, digital leaders like **Amazon and Walmart are aggressively deploying AI shopping assistants**. Amazon’s new GPT-4 powered assistant “Rufus” is trained on its entire product catalog plus web data to answer shoppers’ questions, compare products, and make recommendations in a conversational flow ¹² ¹³. Similarly, Walmart’s AI assistant “Sparky” now lives in the Walmart app, where it **helps customers search, get personalized suggestions, and even plan projects or meals with items added to cart** ¹⁴ ¹⁵. These retail giants see AI agents as *“true AI-powered retail companions”* that simplify shopping and deepen customer loyalty ¹⁶. Zillow provides another example: it built the first real estate plugin for ChatGPT, enabling home shoppers to ask natural questions and receive live listings with photos and pricing – a frictionless experience that *“brings real listings and housing insights into everyday conversations”* ¹⁷ ¹⁸.

By contrast, **legacy retailers and brands that rely on traditional browsing and search traffic face grave challenges**. Department stores like Macy’s, for instance, have struggled to build robust e-commerce and personalization capabilities. If consumers bypass websites in favor of AI assistants, a retailer without a strong AI strategy or integration into these agent ecosystems could see sharply reduced visibility. We argue that businesses stuck in the Web 2.0 mindset – dependent on SEO, web storefronts, and social media ads – will lose ground to those who build AI-native customer experiences. The paper highlights warning signs from current trends: for example, **marketers are already losing share of voice in the discovery phase** as generative AI handles more query answering, diminishing the impact of brands’ search engine marketing ¹⁹.

- **Websites for Humans vs. Websites for Machines:** As the locus of interaction shifts from human browsers to AI agents, the very design of digital content must adapt. We explore the technical contrasts between sites optimized for human viewing and those optimized for consumption by LLMs. Traditional web design focuses on visual appeal and user experience, whereas **LLM-oriented design prioritizes structured, machine-readable content** that can be ingested and synthesized. For instance, AI systems often **extract specific chunks of content to answer questions, rather than reading a whole page** ²⁰. This demands a new approach to content management: *“content-block optimization”* instead of page-based SEO ²¹ ²². We discuss emerging best practices such as ensuring critical information is present in raw HTML (since most AI crawlers do not execute JavaScript) ²³ ²⁴, using semantic markup, and even new conventions like `llms.txt` **files to flag high-value content for AI crawlers** ²⁵. We also introduce the concept of **Agentic Web Interfaces (AWIs)** – a proposed new layer of web infrastructure designed specifically for AI agents to interact with, separate from the human-facing UI ²⁶ ²⁷. In effect, businesses may need to maintain a dual

stack: one experience for human users and another (an API or structured data feed) for AI agents. This dual-stack design ensures that an AI can seamlessly query inventory, prices, or FAQs and execute transactions without relying on scraping a graphical website meant for people.

- **Emerging UX Expectations in Conversational Commerce:** The LLM interface not only changes *what* users do, but *how* they do it. We examine the emerging user experience and design principles in an agent-driven world. One major change is a return to **conversational, personalized assistance** – at scale. Whereas the traditional e-commerce journey might be self-service (browsing menus and filters on a site), conversational commerce feels more like interacting with a knowledgeable sales assistant. Users can ask an AI agent highly specific or open-ended questions (“I need a gift for a 5-year-old who loves dinosaurs” or “How do I fix my leaky faucet?”), and the agent will interpret context and provide tailored solutions ²⁸ ²⁹ . This resembles the high-touch “white glove” service of in-store shopping, now delivered via AI ³⁰ ³¹ .

User expectations will shift toward **natural language engagement, instant answers, and proactive help**. AI assistants can handle follow-up questions, remember preferences, and even anticipate needs. For example, Walmart’s Sparky can suggest a week of meal plans when asked “What’s for dinner?”, automatically adding the ingredients to the cart ²⁹ . Such experiences blur the line between shopping and advisory services. Design-wise, interfaces will favor **chat-centric layouts, voice interaction, and rich media embedded in AI responses**, rather than conventional point-and-click navigation. There is also a greater emphasis on **trust and transparency** – since an AI agent is an intermediary, users must trust it to act in their best interest. The paper discusses how *“advertising becomes the interface”* in some AI scenarios, raising concerns about covert influence ³² ³³ . To maintain user trust, agent-based platforms will need robust disclosure and ethical guardrails (for instance, clearly indicating sponsored suggestions, and requiring user confirmation for purchases, as OpenAI’s guidelines for ChatGPT’s shopping feature do ³⁴ ³⁵). In short, the winning UX in the agentic era is **conversational, context-aware, highly convenient, but also transparent and user-controlled**.

- **GPTs and AI Agents as the New Payment Rails:** One of the most radical implications of LLM-based interfaces is their potential to become a **universal transaction layer**, handling payments and orders across the web. We argue that GPT-powered agents will evolve into payment conduits akin to Apple Pay or PayPal, seamlessly facilitating purchases within the chat interface. Evidence of this trajectory emerged in late 2025: OpenAI introduced an “Instant Checkout” feature in ChatGPT that allows users to **buy products directly in a chat conversation** ³⁶ . Through a partnership with Stripe, ChatGPT can now securely pass order and payment details to merchants (starting with Etsy and soon millions of Shopify stores) so that a user can go from query to confirmed purchase in a few clicks – *“without ever leaving the chat,”* as OpenAI describes ³⁵ . The underlying Agentic Commerce Protocol keeps merchants as the seller of record but lets the AI agent coordinate the transaction and payment on the user’s behalf ³⁷ ³⁸ . In practice, this means the AI agent holds the user’s payment credentials (encrypted and permissioned) and can act as a **universal checkout**.

The implications are far-reaching. Just as stored-payment systems like Amazon 1-Click and Apple Pay boosted conversion rates by removing friction, AI agents handling payments could greatly streamline online commerce. A user might tell their assistant, “Order me the top-rated air purifier under \$300,” and the agent will find the product, confirm the choice, and complete the purchase instantly using stored payment info. This flips the traditional model – instead of navigating to a specific merchant’s site or app, the *agent* orchestrates the purchase through whichever merchant can fulfill it best (much like a personal shopper). Payment providers are actively embracing this future: PayPal, for example, has struck a deal to integrate its

service into ChatGPT, seeing the agentic interface as an important new channel for transactions ³⁹ ⁴⁰ . Over time, one can imagine AI platforms becoming **payment hubs**, charging small facilitation fees per transaction (as OpenAI does in its pilot) rather than relying on advertising revenue. Our thesis is that by 2030, GPT-like agents will be as routine for executing payments and orders as credit card terminals and payment apps are today.

- **Decline of Search Ads, SEO, and Traditional Digital Marketing:** With AI agents mediating so many interactions, the established pillars of digital marketing will need to be rethought. We present evidence that **search advertising and traditional SEO are already losing effectiveness** in the face of generative AI. Marketers have long optimized content to rank in Google results and paid for AdWords to capture search traffic. But when an AI agent provides a direct answer or recommendation, it often bypasses the list of blue links (and their ads). Bain's research highlights that *about 60% of searches now end without a click* to any external site ⁴¹ , and this zero-click phenomenon is accelerating across demographics. As a result, companies can no longer count on users visiting their homepage or landing page – the decision may be made within the agent's answer. This reduces the value of investments in content marketing and diminishes the ROI of search ads if fewer eyeballs ever see them.

Social media marketing will face similar headwinds. Today, many consumers discover products via Instagram feeds, TikTok videos, or influencer recommendations. In an agentic web, however, a user might simply ask their AI for product ideas, and the AI will deliver suggestions drawn from its training data and real-time databases – potentially sidestepping the social discovery process. In other words, **AI agents compete for the user's attention and purchase intent, potentially displacing some of the influence of social media ads and sponsored posts**. Moreover, AI systems personalize recommendations deeply to the user's stated needs, which could reduce the impact of broad-spectrum advertising. We discuss how Meta and others are adapting – for example, Meta has reportedly explored using generative AI chat data to better target ads on its platforms ⁴² , an attempt to exploit the “intimacy” users have with AI conversations. Yet this raises ethical and practical issues, as users may react negatively if they feel their private AI dialogues are mined for ads. Regulators are already scrutinizing the possibility of covert advertising in AI assistants ⁴³ ³² .

Overall, the *value of traditional impressions and clicks is likely to erode* in the LLM era. A16Z analysts note that the classic search engine monetization model (users pay with attention to ads) is giving way to a model where many LLM services are subscription-based ⁴⁴ . This “*structural shift*” means **LLM platforms have less incentive to drive users to third-party sites** and may only do so when it truly enhances the user experience ⁴⁵ . In fact, an AI platform might prefer to answer a query with an in-line quote or summary from a source (satisfying the user immediately) rather than send the user off-site – the opposite of how search engines optimized for ad clicks. This suggests that **content visibility will be measured in citations and references within AI answers (“reference rate”) rather than page views** ⁴⁶ . Marketers will need to master what has been dubbed **Generative Engine Optimization (GEO)** – influencing how AI models perceive and present their brand and information ⁴⁷ ⁴⁸ . Later sections provide insight into strategies for this new form of optimization.

- **Monetization Opportunities in Building AI Agents and Platforms:** A bright spot in this disruption is the emergence of entirely new markets and revenue streams centered on LLM technology. We explore how businesses and individual developers can create value in the “**agent economy**.” One opportunity is building custom AI agents (or “custom GPTs” and “Gems,” in OpenAI and Google's terminology) tailored to specific domains or company data. OpenAI opened up a GPT **Store** in 2024

where developers can publish custom versions of ChatGPT for others to use, with plans for a revenue-sharing model based on usage ⁴⁹. In essence, this is the beginnings of an **AI agent marketplace**, analogous to an app store. Already millions of custom GPTs have been created in the first months of that launch ⁵⁰ – ranging from writing tutors to cooking assistants – pointing to a vibrant ecosystem. Google's Gemini LLM similarly introduced "Gems," which empower users to create their own mini-AI experts for various tasks ⁵¹ ⁵². We argue that companies should view these custom agents as the next frontier of customer engagement and even a product offering in their own right. For example, a bank might deploy a fine-tuned financial advisor bot, or a health provider offers a custom GPT trained on its medical knowledge. These agents can deepen customer lock-in and potentially be monetized via subscriptions or upsells.

Beyond direct consumer-facing agents, **enterprise use-cases for LLM agents are poised to boost productivity and spawn new enterprise software solutions**. Microsoft's early data shows huge interest from organizations in building internal copilots – more than 230,000 organizations had used Microsoft's tools to create AI agents or automations by 2025 ⁵³. These range from sales-lead prioritization bots to healthcare documentation assistants ⁵⁴ ⁵⁵. The enterprise AI agent market could mirror the explosion of SaaS in the 2000s, with consultancies and startups offering fine-tuned models for every vertical (legal, HR, education, etc.). There will also be demand for infrastructure that supports agents – e.g., systems to manage "agent sprawl," ensure security and compliance of automated decisions, and measure agent performance ⁵⁶ ⁵⁷. All of these are budding monetization avenues.

In short, while some traditional online revenue streams (like display ads or SEO consulting) may shrink, the **LLM revolution creates new ones: selling AI expertise, custom models, API services, and AI integration tools**. We liken this to the mobile app boom – companies that quickly built valuable apps gained millions of users; now, those who build the most useful and trustworthy AI agents may become the next generation of powerhouses (or at least add significant new revenue lines).

- **Timeline and Trajectory (2025–2030):** The paper offers a forecast for how quickly agent-based commerce could become dominant and benchmarks to watch for. Based on current adoption curves, we predict that **2027–2030 will be the period when AI-agent interfaces move from early adoption to mainstream dominance in digital commerce**. This aligns with independent forecasts that by 2030, a majority of information queries globally will be handled by AI agents ⁹ ⁵⁸. Milestones on the way to that endpoint include: widespread integration of AI into all major consumer devices (e.g. Apple's likely moves to deeply embed LLM assistants into iOS, following their 2025 support for AI search in Safari ⁵⁹), continued doubling of voice-assistant usage (which numbered 8.4 billion devices in use by 2025 ⁶⁰, and will increasingly be LLM-powered), and the refinement of multimodal agents that can see and act (e.g. tools that can handle images, voice, and text commands fluidly by 2027).

We address the comparison to past hype cycles, notably the Y2K phenomenon and the dot-com bubble, to temper the timeline with realism. Unlike Y2K – a hard deadline that led to massive investment but ultimately averted disaster – the rise of AI agents is not a single moment but a progression. There may well be hype-fueled disappointments along the way (as inflated expectations in 2023–2024 get recalibrated), but the structural advantages of agentic interfaces (convenience, efficiency, personalization) ensure that adoption will keep rising. If anything, early hype might lead some to dismiss LLMs as a fad if the impact isn't immediately catastrophic to the old guard. We caution that such thinking is dangerous for businesses. The *real disruption* often unfolds a few years after the hype peak – just as the dot-com bust in 2000 was followed by the true web revolution of the 2000s. By the late 2020s, the **agent-based paradigm will likely be in its**

“inevitable” phase, where consumers simply expect to accomplish tasks by telling their AI what they want. Companies that have not prepared for that reality by 2027 could find themselves as irrelevant as those that failed to build a web presence by the late 1990s.

- **Strategic Guidance for Businesses:** Finally, this thesis offers frameworks and recommendations for organizations to **future-proof themselves for LLM-driven commerce**. Key strategies include: (1) **Developing a dual content strategy** – one that continues to serve human visitors with rich experiences, and another that ensures machines (AI crawlers and agents) can easily find, parse, and utilize your data. This may involve publishing structured data feeds, adding semantic annotations, and embracing standards like schema.org and `llms.txt` to signal important information ²⁵. (2) **Integrating with AI platforms** – e.g., making your products or services available through plugins or APIs to popular assistants (as early adopters like Instacart, Expedia, and Zillow have done with ChatGPT). If there are emerging marketplaces for agents (OpenAI’s GPT store, Microsoft’s Bing Chat plugins, Google’s Gems), treat them with the same priority that you treated app stores or search engines in the past – they are the new discovery channels. (3) **Building your own agents** where appropriate. If you have a substantial user base or data advantage, consider offering an AI assistant feature yourself. This can be internal (to augment your customer service or sales) or external (a new product feature for users). For example, banks are creating chatbot advisors, and retailers are embedding AI in their apps to guide purchases. Even if you don’t have the resources of an Amazon or Walmart, you can leverage third-party LLMs via APIs to power these features. (4) **Rethinking marketing and SEO for the AI era**. This means monitoring not just web analytics but also how your brand is mentioned in AI responses. New tools are emerging for *AI visibility* – such as platforms that analyze which questions your brand appears as an answer for ⁴⁶ ⁶¹. Marketers should experiment with prompt engineering: e.g., publishing Q&A content that aligns with how users might query an AI, or ensuring your brand’s factual data (prices, specs, locations) is up-to-date on knowledge graphs that LLMs train on. Additionally, branding and trust-building become even more crucial – if an AI agent tends to recommend one or two options for a given query, brands need to earn that recommendation through superior offerings and customer satisfaction (since algorithmic favoritism will likely stem from user ratings and relevance). (5) **Investing in data and AI infrastructure**. Companies must treat their data (product data, customer interaction data, etc.) as a foundation for AI. Clean, rich data will enable better AI-driven services. Also, consider fine-tuning or training smaller LLMs on your proprietary data where needed, so you aren’t entirely dependent on generic models that might not know the nuances of your business. (6) **Navigating ethical and regulatory aspects**. Be proactive about the privacy and fairness implications of AI commerce. Transparency with users – e.g., disclosing when an AI is used, or explaining how an AI made a recommendation – can build trust. Also, plan for regulations (like the EU’s AI Act) that might require documentation of AI systems’ behavior ⁶². By building ethical principles into your AI deployments now (such as avoiding manipulative designs ⁴³), you both future-proof against legal risk and differentiate on trust.

The overarching guidance is that **businesses should embrace a mindset of “co-designing” their future with AI**. Much as companies in 1995–2000 learned to design websites and optimize for search engines, today they must design processes and customer journeys that include AI agents as first-class participants. This might mean **redesigning e-commerce flows** so an AI can traverse them end-to-end – for instance, providing an API for adding items to a cart and checking out, which an agent can use directly (as opposed to a GUI-only workflow). It also means training staff to leverage AI tools internally (augmenting human decision-making with AI suggestions). Organizations that cultivate an AI-savvy culture and experiment rapidly will be more prepared for the agent-dominated market by 2030.

In conclusion, **the rise of GPTs and AI agents represents a paradigm shift in digital commerce and user interfaces** on par with the advent of the web itself. The familiar model of opening a browser, typing queries, and navigating websites is being replaced by fluid conversations with intelligent agents that *know* the user and can *do* on the user's behalf. This transition, the heart of the LLM Interface Economy, is not hype but an observable, data-backed trend that is already changing consumer behavior. Its impacts will reverberate across every industry: retailers will compete to be the preferred recommendation of AI assistants, marketers will optimize for AI over search, and consumers will come to expect “one-stop” assistance in their digital lives. Companies that recognize the *methodical inevitability* of this shift – and act with vision and pragmatism – have the opportunity to secure their place in the next era of commerce. Those that ignore it may wake up to find that by the time they build their “AI presence,” the crowd (or rather, the AI agents guiding the crowd) has moved on. The time to adapt is now. As one technologist aptly put it, in the near future “*we will not surf the web; we will converse with it*” ⁶³ – and those conversations will drive the global economy.

(The remainder of this document is organized into chapters that explore each of these points in depth, complete with case studies, technical analysis, and cited evidence from industry research. We begin in Chapter 1 by charting the historical evolution of web interfaces, laying the groundwork for why this moment is akin to the inflection points of 1994 and 2007.)

1. From Web 1.0 to Web 4.0: A Brief History of Interface Paradigms

Web 1.0 – The Era of Domains and Direct Navigation (1990s): The first era of the web (roughly 1991–2000) introduced the basic architecture of online commerce and information – the website. In Web 1.0, content was largely static and read-only. Users discovered sites through directories (like Yahoo) or by typing in known **domain names**, which quickly became prized assets for businesses (much as a good storefront location is valuable in the physical world). The mid-1990s saw explosive growth in websites, and companies realized that a memorable `.com` address and inclusion in web directories or portal listings were critical for consumer discovery. **Domain names were transformative circa 1994–1995:** obtaining a simple domain gave companies a direct channel to reach the then-small but rapidly growing online audience. In 1994, there were fewer than 3,000 websites; by 2000, there were over 25 million – a testament to the rush to establish a web presence. Commerce began moving online as pioneers like Amazon (founded 1994) and eBay (1995) built businesses that were “*born digital*”. Still, the user experience in Web 1.0 was often described as “surfing” – manually hopping from page to page via hyperlinks or URL entry, with minimal personalization or interaction.

Web 2.0 – Search, Social, and Mobile (2000s to mid-2010s): The early 2000s brought a shift to more dynamic, user-driven content. **Search engines**, especially Google (founded 1998), became the new gatekeepers of the web. Instead of curated directories, algorithms indexed the exploding number of pages and answered user queries with ranked links. This gave rise to the field of **Search Engine Optimization (SEO)** – businesses invested heavily in techniques to ensure their sites ranked highly on Google's results ⁴⁷. The importance of search in this era cannot be overstated: it was the primary way users found information or products, effectively making Google the world's homepage. At the same time, Web 2.0 saw the emergence of **social media** and user-generated content (Facebook, 2004; YouTube, 2005; Twitter, 2006). These platforms changed digital commerce by enabling word-of-mouth at scale and new forms of advertising (social ads, influencer marketing). Later, the **smartphone revolution (post-2007)** added another layer: mobile apps created an alternate ecosystem alongside the browser. The late 2000s and early 2010s saw businesses scrambling to launch apps and adapt to users spending more time on mobile devices

than desktop. This era's hallmark was multiple, siloed interfaces – one might use a search engine to find a product, then a retailer's website (or app) to buy it, and perhaps a social network to review or share it. Companies had to optimize across **web, mobile, and social channels simultaneously**. Many succeeded (e.g., Amazon leveraged both SEO and its mobile app, while building its own search-like product recommendation engine internally). Others struggled – for instance, some traditional retailers never mastered SEO or the shift to online, contributing to the decline of malls and brick-and-mortar incumbents.

Web 3.0 – The Semantic and Decentralized Web (late 2010s): The term “Web 3.0” has been used to describe a few intertwined concepts in the late 2010s and early 2020s. One aspect was the idea of a **semantic web**, where content would be structured in ways that enable machines to understand context and meaning (e.g., the use of structured data markup so search engines can directly display answers, like rich snippets). Another aspect – more hyped in 2020–2022 – was the **decentralized web or “web3”**, involving blockchain, cryptocurrencies, and user-owned networks. Proponents imagined a web where users had more control over their data and identity, and where value could be transacted peer-to-peer without centralized intermediaries. While decentralized platforms did not see mainstream consumer adoption in commerce (and by 2023 the crypto/Web3 hype had cooled), the semantic web did quietly advance. Search engines and voice assistants relied increasingly on structured data and AI to serve direct answers. For example, Google's Knowledge Graph (introduced 2012) and rich results meant users often got instant answers or could complete tasks (like booking a flight or getting a sports score) **without visiting a website**, an early sign of the zero-click trend.

Web 4.0 – The Agentic Web (mid-2020s and beyond): We stand at the dawn of what can be called Web 4.0, where the central paradigm is **AI agents orchestrating interactions**. If Web 3.0 was about machine-readable data, Web 4.0 is about machine-actionable tasks. In this phase, *agents become first-class actors on the web*. A user's personal AI might negotiate with another AI (say, a shopping bot interacting with a merchant's inventory system) to get things done. The interface is conversational and contextual rather than page-based. Notably, **the Agentic Web does not discard the gains of previous eras but builds on them**: it leverages the vast data (including semantic markup) accumulated on Web 3.0, it still uses the connectivity of Web 2.0 (APIs, cloud services), and it inherits Web 1.0's fundamental protocols (HTTP, etc.). But it *reshapes* how we experience these elements – the browser and search box give way to an omnipresent assistant embedded in our devices and apps.

One can draw analogies to past transitions. For instance, in the AOL-to-Open-Web shift: AOL in the 90s provided a walled-garden, human-curated experience. When people ventured to the open web, it was chaotic but vastly richer, and open search engines became essential to navigate it. Similarly, today's web is a sprawling universe of content and apps; AI agents promise to tame this complexity by **“pulling the terrain toward the user”** with each query, as one observer described – meaning the agent gathers and composes the relevant pieces of the web for you, rather than you traversing the web yourself ⁶⁴ 4 . Another historical parallel is the mobile app explosion: in the late 2000s, many predicted that apps might kill the open web as people prefer the smoother UX of apps. To an extent, apps did change user behavior, but they did not eliminate the web. Instead, we got a dual system (apps for frequent interactions, web for infrequent or deep information). Now AI agents might abstract over both apps and web. For example, instead of using ten different apps (one for rideshare, one for food delivery, one for shopping), a user might just tell their AI, “I need a ride to the airport and a sandwich on the way,” and the AI handles coordinating multiple services behind the scenes. This indicates **a unification of services at the interface level** – the user talks to one AI which in turn can utilize many back-end services.

Importantly, Web 4.0 is not fully “here” yet in 2025 – but the pieces are rapidly falling into place. The next sections detail those pieces: the rise of generative AI usage, the changes in consumer behavior, and the strategic moves by major tech companies that all point to an agentic future.

(Citations for this section: Historical web evolution points are drawn from known timelines and industry retrospectives. The transformative role of domain names in the 1990s is well documented in internet history archives. The growth of search and social in Web 2.0 is evidenced by Google’s dominance in online ad share and the emergence of SEO as a \$80+ billion industry ⁵⁹. The shift toward zero-click answers was noted by Bain’s 2024 consumer survey ⁸, and Apple’s integration of AI into Safari by 2025 was reported by Andreessen Horowitz analysts ⁵⁹. The term “agentic web” and the concept of agents acting on behalf of users is supported by Microsoft’s 2025 vision statement ³.)

2. The End of Browsing: How LLMs Are Changing User Behavior

In 1996, Steve Jobs likened the web browser to a “reading machine” – a tool people use to read and navigate the vast library of information online. For decades, that metaphor held true. But now, in the mid-2020s, the very act of **“browsing” the web is quietly collapsing into something new**. This chapter examines how **user behavior is shifting away from manual browsing and searching toward conversational inquiry and direct task completion**, driven by improvements in LLM capabilities. The data and early trends strongly suggest that we are witnessing the beginning of the end for the browser-centric way of doing things. As one technologist poetically observed, *“The old web is still alive, but it walks like a ghost... We used to go somewhere to find things. Now the thing comes to us, summoned by language.”* ⁶⁵.

2.1 From Surfing to Summoning: Users Prefer to Ask, Not Search

The hallmark of the traditional web experience was **active user navigation** – typing keywords, clicking links, exploring sites. It was often described as *“surfing”* an ocean of information. In contrast, the emerging mode is **“summoning”** information on demand. Instead of skimming many sources and assembling an answer, users can now pose a question or task to an AI and receive a synthesized response or action. This is a qualitatively different mental model.

Evidence of this shift appears in usage statistics of AI assistants versus search engines. In late 2022, when ChatGPT was first released, it gained 1 million users in 5 days and 100 million within two months – the fastest adoption of any consumer application at the time ⁵. By early 2024, ChatGPT’s web traffic had reportedly overtaken Bing’s, and by some measures it became the second most-used “search tool” after Google ⁶⁶ ⁶⁷. By April 2025, ChatGPT had around **800 million weekly active users and was handling 1 billion+ queries per week** ⁶. These are astounding figures for a product that did not exist prior to late 2022. They indicate a real consumer hunger for this new interface.

Crucially, surveys show that people use ChatGPT and similar LLM bots *for many of the same purposes they’d use a search engine*. A U.S. survey in March 2025 found **52% of adults had tried an AI LLM tool** like ChatGPT, and among those users, about **66% say they use it “like a search engine” for information retrieval** ⁷ ⁶⁸. Top use cases included researching a topic, getting explanations, and even shopping advice. In fact, around **42% of LLM users in late 2024 said they ask AI platforms for shopping recommendations** ⁶⁹ – a clear sign that AI is encroaching on territory once dominated by search engines, retail websites, and social media recommendations.

It's important to note that this doesn't mean users have completely abandoned traditional search. Many reports, including the cited survey, indicate that people still use Google for certain things and AI for others – often depending on the complexity of the task ⁷⁰ ⁷¹. For example, for a simple factual query (“What’s the capital of Bulgaria?” or “weather tomorrow”), users might trust the quick Google answer or a phone widget. But for **open-ended or multi-step tasks** (“Plan a 5-day trip to Bulgaria with kids” or “Compare the best air purifiers under \$300”), users are increasingly leaning on AI assistants which can provide a nuanced answer or a curated set of options in one go ⁷² ⁷³. This division of labor has been observed anecdotally and in user studies: one analysis noted **Google is still used by 98% of ChatGPT users, but those users allocate “the right tool for the job” – Google for quick hits, ChatGPT for detailed synthesis and advice** ⁷⁰ ⁷⁴.

The net effect is that **some portion of traditional search traffic is being siphoned off by LLMs**, and that portion is growing. Gartner predicted that by 2026, traditional search queries might drop by 30% due to AI chatbots and assistants intercepting them ⁷⁵. TTMS, a technology analysis firm, forecasts that **around 2028–2030, LLM-based search will overtake classical search in volume** ⁹ ⁵⁸. Their model, based on current growth rates, suggests AI assistants could handle *over 50% of global search query volume by 2030* ⁷⁶ ⁵⁸. They even cite an analysis by Kevin Indig projecting that ChatGPT’s traffic could rival Google’s by October 2030 if trends hold ⁹ ⁷⁷. While such long-range forecasts have uncertainty, the consensus is that the **trajectory favors AI**. Consumers, especially younger ones, are developing “AI-first” habits. Nearly 80% of Gen Z has used generative AI tools, and a majority find it easier to get information that way – they’re comfortable asking chatbots for **homework help, product ideas, or life advice** in a manner older users might not be ⁷⁸ ⁷⁹.

Another channel shift is happening via **voice assistants integrating LLMs**. By 2025, there were an estimated **8.4 billion voice assistant devices in use globally** (phones, smart speakers, cars, etc.), and roughly 20–30% of consumers use voice search regularly ⁶⁰. These voice assistants (Siri, Alexa, Google Assistant) historically had limited capabilities and often just read out web search results. But as they incorporate advanced LLMs (something all the major players are now pursuing), they effectively turn into on-the-go AI concierges. Asking **“Hey Siri, find me a good Italian restaurant nearby and book a table”** might leverage an LLM to interpret preferences (what is “good?”), search data, and an OpenTable API to complete the task – none of which requires the user to ever open a browser or look at a screen. In short, the **frictionless, conversational style of obtaining information is becoming normalized**, whether via text chat or voice.

To put it succinctly, **users are transitioning from an active search paradigm to an interactive dialogue paradigm**. They pose objectives and questions, and expect the system (composed of AI plus underlying services) to handle retrieval and execution. This is the essence of “summoning” – the user’s intent gravitationally pulls the needed data and actions to them ⁶⁴. The better AI gets, the more users trust it with bigger requests, creating a positive feedback loop (more usage leads to more improvement and investment, which leads to more usage).

2.2 Diminishing Role of the Traditional Web Browser

If information and transactions come to the user via an AI interface, what does that imply for the web browser – that all-purpose tool for navigating websites? While it’s too early to proclaim the browser “dead,” we can see signs that its role is contracting, or at least changing form.

First, consider **where digital interactions begin**. It used to be common to start with a web browser for almost anything on a computer. Mobile changed that somewhat by pushing users to start in an app. Now, AI is altering it again: interactions might start with a voice command or in a chat box (which could be within any app or OS, not necessarily a “browser” app). For instance, Windows 11 now integrates **Windows Copilot**, an AI assistant, right in the taskbar for users to query instead of opening a browser and searching. Similarly, smartphones might soon have a native AI assistant interface that handles tasks without launching Safari or Chrome. So the **entry point is shifting away from the browser UI**.

Second, we have the phenomenon of **“context engines” replacing manual navigation**. A context engine is essentially an AI that can build a temporary workspace of information around your query ⁸⁰. For example, if you ask an AI, “I’m thinking of buying a hybrid SUV, what should I consider?”, the system might behind the scenes fetch consumer reports, spec sheets, reviews, maybe your past car purchase history, and then give you a coherent answer or a set of options – almost like it assembled a mini-website or briefing for you on the fly. When it’s done, that context can be discarded. In such a scenario, you never manually browsed any site, though the AI likely pulled from many sites. The browser’s function as a container for content from different sources is being usurped by the AI’s ability to *aggregate content into a single answer*.

A striking way to phrase this (as in the source excerpt above) is: **“The browser becomes a museum interface – a preserved shell for static artifacts”** ⁴. That is, traditional websites and HTML pages might persist, but mainly as archives or specialized tools, not the default way most users get things done. People might visit a browser for niche needs – e.g., if they want a very specific webpage or to verify an AI’s answer by reading the source document in full – but these could become occasional activities. The everyday actions (finding info, shopping, communicating) could all be mediated by AI through simpler interfaces.

Already, we see declining web traffic in areas where AI provides alternatives. A concrete example: **Wikipedia’s traffic has dropped in recent years**, partly because search engines like Google directly show snippets from it (and now AI bots just answer using its content without the user clicking through). When an AI like Bing’s or Bard’s can answer “What is quantum physics?” with a multi-paragraph explanation sourced from top encyclopedias, a user may not feel the need to open Wikipedia at all. Multiply that by millions of topics, and you reduce a significant chunk of traditional browsing.

Another angle is **multi-step tasks**. Let’s say you want to plan a weekend trip. The traditional way: search for “weekend trip ideas”, click some travel blogs, search for hotels on Booking.com, go to a map site for routes, maybe visit a weather site, etc. That’s multiple browser tabs and manual comparisons. The AI-driven way: you tell an agent your preferences (“Find a quiet mountain town within 5 hours drive, pet-friendly, with a spa, and book a top-rated 3-star hotel for two nights”) – the agent might consult various APIs and sites, then present you with a plan and ask for confirmation to book. Perhaps it shows a few options to choose from, but far fewer than the dozens of blue links you’d wade through on your own. **This shift compresses the user journey**. One could say the *entire travel booking workflow becomes one “conversation” with the AI* rather than dozens of page loads and form fills. When you do this via an AI chatbot (whether in a messaging app, a dedicated AI app, or voice assistant), you might not even realize a browser is involved. Technically, the agent might use behind-the-scenes browser automation or APIs, but the user isn’t engaging with a browser UI.

It’s telling that tech companies themselves are preparing for this. Google, whose Chrome browser is a linchpin of the traditional web, is now integrating **generative AI answers in its search results (SGE – Search Generative Experience)** to keep users from going elsewhere. Sundar Pichai, Google’s CEO, in late 2024 even acknowledged that the familiar search page will evolve radically, implying that the era of 10 blue

links is ending. As mentioned, Google's search head Elizabeth Reid said the *search bar will be less important over time as users interact more fluidly with AI* ¹¹. Microsoft, which has a smaller browser share, is nonetheless deeply embedding AI in Edge and Windows to gain an edge (no pun intended) in this new interface war.

There is also an aspect of **attention economics**. As generative AI fulfills more requests directly, **the attention once spent browsing multiple sites is recaptured into the AI's interface**. This benefits the AI platform – for example, users might spend longer in a ChatGPT session exploring various follow-up questions (OpenAI reported multi-turn conversations average several queries). For the user, it's arguably a win in efficiency (less time hunting for the right page), but it does mean fewer opportunities for websites to grab that attention. It calls to mind Bain's finding: *80% of consumers use AI answers frequently, resulting in 15–25% less organic traffic to websites overall* ⁸. Over time, if these trends continue, companies that rely solely on people coming to their site via a browser will feel the pinch.

Does this mean the web and browsers vanish? Not entirely. Web pages still serve as the **source material** for a lot of AI outputs. Many AIs, including open models, are trained on the web's corpus. And for content that is richly visual or interactive (e.g., detailed dashboards, design tools, immersive experiences), the browser will remain useful. Also, professionals and researchers might still use browsers to access raw information or specialty databases. However, for the **mass consumer use cases** – the everyday questions and chores – the browser may fade into an intermediary that one uses occasionally, while the default is to ask the agent.

An analogy: Think of the command line vs. GUI on computers. The command line interface (CLI) is still there on every OS, and power users still use it, but most people default to the graphical user interface (GUI) because it's easier. In our case, the old style (manually browsing) might become akin to the CLI – powerful but only used when needed – whereas the AI chat is the friendly GUI for most tasks.

Some tech commentators have framed it as **"Browsers become the new back-end"** – meaning the browser might run invisibly under the hood, or render content for an AI to scrape, but human users won't engage with it directly as much. Already, AI crawlers are voraciously reading web content (OpenAI's GPTBot, for instance, was reported to consume *30 TB of data from a single site in a month* as it trained ⁸¹). It's as if the AI is becoming the primary "user" of websites, reading them to serve many end-users indirectly.

We also see early product decisions reflecting a diminished browser role. OpenAI's ChatGPT introduced **plugins and the ability to browse** (with Bing) so that it could fetch live info itself. In those cases, the AI is using a hidden browser to get data, but the user just sees the answer. Another example: Amazon's Alexa and Google Assistant can now complete some orders or find info without sending you a link – they'll just voice out the result.

In summary, the traditional browser isn't going to disappear overnight, but its **centrality is waning**. As AI agent adoption grows, the fraction of user problems solved *without opening a browser* will keep increasing. It is quite plausible that by 2030, a significant majority of routine queries (information lookup, basic shopping, booking, etc.) will not involve a user navigating a website at all – they will be handled start-to-finish by agentic interactions. The browser might persist for *"long tail"* use cases and certain complex interactions, much as mainframe terminals persisted even as PCs became mainstream, but it will no longer be the default doorway to the digital world. In the words of one manifesto on this topic, *"The browser will die quietly"* ⁸² – not with a sudden bang, but by gradually being relegated to a supporting role while AI agents become the star.

2.3 Implications for E-Commerce and Online Business

For businesses, the decline of browsing and rise of asking carries significant implications. When users stop leisurely browsing storefronts or Googling around for options, the traditional methods of attracting and converting customers must be rethought. Some key implications include:

- **Loss of Direct Traffic and Discovery:** Companies historically invested in making their homepage or app a destination – via branding, bookmarks, emails (“Visit our website for deals!”), etc. If users just ask an agent for, say, a product recommendation, they might never see the company’s beautifully designed homepage or promotional banners. Instead, the agent might present a generic description or a comparison. This means brands could lose some ability to *shape the customer’s journey visually or emotionally*. They’ll need to ensure their product information and reputation speak for themselves because the AI will likely choose on objective grounds (price, ratings, fit with user query).
- **Compression of the Funnel:** The marketing funnel (awareness → consideration → decision) can be collapsed by an AI that provides awareness (by presenting a product the user didn’t specifically name), does the consideration (by summarizing reviews or specs), and drives decision (by facilitating purchase) all in one chain of interactions. The positive is it could lead to faster conversions (fewer opportunities for the customer to get distracted or abandon the process). The negative for businesses is *less opportunity to intercept or influence* at each stage. The AI might not show the top 10 results (which you could pay to be in via ads); it might just say “I recommend Product X.” If you’re not Product X, you don’t even get a chance to put up a pop-up or offer a last-minute coupon to sway the customer.
- **SEO and Content Strategy Overhaul:** As noted earlier, being on page 1 of Google may matter less than being mentioned by AI. Companies will likely need to adopt **Generative Engine Optimization (GEO)** strategies ⁴⁷ – optimizing content so that it is *selected and accurately used by AI models*. This could involve providing data feeds to AI providers, using structured data so the AI can easily quote your content, and focusing on *authority and clarity* in content. For instance, an FAQ or guide on your website that is succinct and well-structured is more likely to be picked up verbatim by an LLM (which prefers “**content that is well-organized, easy to parse, and dense with meaning**” over SEO-style keyword stuffing ⁸³ ⁸⁴). As A16Z’s report stated, *phrases like “in summary” or bullet points help LLMs extract content effectively* ⁸³ ⁸⁵ , suggesting that how you format information for AI consumption might differ from human-facing copywriting.
- **New Metrics of Success:** Instead of click-through rates, marketers might look at “**reference rates**” – how often their brand or pages are cited in AI answers ⁴⁶ . There are already startups and tools aiming to measure this. For example, some services can query AI models with various prompts to see if/when a brand comes up, effectively creating an “AI search share” metric. Traditional web analytics (page views, time on site) will partly give way to *AI analytics* – like tracking if the AI’s summary of your product is correct and positive, or if users gave feedback (via the AI interface) that they were satisfied with the recommendation.
- **Greater Emphasis on Reviews and Quality Signals:** AI recommendations will heavily rely on aggregated data like customer reviews, ratings, and third-party evaluations to choose products or answers. If a user asks, “What’s the best budget smartphone?”, the AI will likely parse specifications and user ratings across the web to pick an answer (especially if not explicitly trained on proprietary

data of any one manufacturer). Thus, companies must ensure their products actually perform well and satisfy customers, because no amount of marketing spend can easily overcome a facts-driven AI that sees *Product A has a 4.7 star rating and Product B has 4.2 – it will probably suggest A* (unless the user specifies some other priority). **In an AI-mediated world, the manipulable surface (like ad copy or shiny packaging) matters less, and objective metrics (quality, price, popularity) matter more.** Some have likened this to a return to *substance over style*, enforced by algorithms that scour for the best data points.

- **Fewer Opportunities for Upselling and Cross-Selling (Unless AI-Integrated):** On a website or app, once you have a customer's attention, you can merchandise related items ("Customers also bought..."). In an AI agent, the agent might handle that role. If it is truly acting for the user's benefit, it might suggest a cheaper alternative if it thinks the user is cost-conscious, or remind them of an accessory they might need. If you are the business, you'd want to ensure the AI has all the info about your complementary products or services so it can do effective cross-sell on your behalf. But if it doesn't, you lose that upsell revenue. This implies providing AI platforms with structured catalogs of your offerings and the relationships between them (something like Amazon's "frequently bought together" data, but accessible to any AI that's shopping).
- **Customer Loyalty and AI Intermediation:** Companies will need to grapple with the fact that **the AI might become the holder of customer loyalty, not the brand or platform.** For example, if people start saying "Hey GPT, order me my usual groceries," the user's loyalty is to the assistant (for convenience), and the assistant might choose whichever grocery retailer is most efficient or cost-effective in the background. If the AI switches vendors because it found a better deal and the user doesn't mind or even know, the traditional notion of customers sticking to a particular store or site diminishes. To counter this, brands might either need to *integrate with AIs* so they are always considered (e.g., ensure your store is a default provider for something in the AI's network), or they need to find ways to give the AI incentives to stick with them (which could be tricky unless some sort of paid partnership or "AI marketing" develops – an area we explore later in monetization).
- **Continued Importance of the Web, but as a Back-End:** The web will still be where a lot of content resides and transactions are executed. But businesses might increasingly treat their web presence as an **endpoint for AIs** rather than primarily for users. For instance, rather than focusing exclusively on a flashy web UI, a company might invest in a robust web API or content feed. E-commerce sites might expose their inventory and prices in a structured format (some already do via Google Shopping feeds, etc.) so that AI agents can query availability. News organizations might allow LLMs to directly pull summarized facts (with proper licensing or attribution) instead of hoping users click through a headline. In essence, websites will need to become **machine-friendly** by design (fast loading, no login walls for basic info, structured content), because AIs have "no patience" for things like excessive scripts or hidden info ⁸⁶ ⁸⁷. Those that fail to do so might be ignored by AI agents that skip slow or hard-to-parse sites ⁸⁶.

In conclusion, the behavioral shift away from browsing and toward asking/conversing is already underway and measurable. It heralds a new competitive landscape where **being the chosen answer** or service in an AI-driven exchange is the paramount goal. Traditional web design and marketing must adapt to this reality. Companies should not panic – the sky isn't falling all at once – but they absolutely should plan for a future where *most customer interactions are mediated by an AI agent*. The subsequent chapters delve into how

companies can position themselves as winners in this future, what new technical and strategic challenges arise, and how to navigate the transition period between the old model and the new.

(Citations in this section support the trends discussed: ChatGPT usage surpassing Bing and hitting 100M+ users ⁵ ⁶⁷ ; majority of LLM users treating them like search ⁷ ; 80% zero-click behavior and traffic drop stats from Bain ⁸ ; Gen Z and overall adoption stats ⁷⁸ ⁶ ; Microsoft and Google acknowledging the changing search interface ¹¹ ; the notion of context engines and the “browser as museum” quote ⁴ ; the GPTBot 30 TB crawl example ⁸¹ highlighting how AIs consume web content directly. These underscore the factual basis for the qualitative changes described.)

[The paper continues with Chapter 3, which examines specific case studies of companies adapting to the LLM interface economy – contrasting proactive innovators like Amazon, Walmart, and Zillow with laggards – and Chapter 4, which dives into the technical side of how websites and content should be structured for AI consumption (dual-stack design, AWIs, etc.).]

¹ ⁴ ⁶⁴ ⁶⁵ ⁸⁰ ⁸² **Part IV — The End of Browsing and the Architecture of Conscious Search**

<https://www.notion.so/a857c007afa942b9be4f794c4a25928e>

² ⁶³ **Part III — Agentic Harmony and the Age of Co-Creation**

<https://www.notion.so/356236b0aa094ba2b0bc92e2c3833459>

³ ⁵³ ⁵⁴ ⁵⁵ ⁵⁶ ⁵⁷ **Microsoft Build 2025: The age of AI agents and building the open agentic web - The Official Microsoft Blog**

<https://blogs.microsoft.com/blog/2025/05/19/microsoft-build-2025-the-age-of-ai-agents-and-building-the-open-agentic-web/>

⁵ ⁶ ⁷ ⁹ ¹⁰ ¹¹ ⁵⁸ ⁶⁰ ⁶⁶ ⁶⁷ ⁶⁸ ⁷⁰ ⁷¹ ⁷² ⁷³ ⁷⁴ ⁷⁶ ⁷⁷ ⁷⁸ ⁷⁹ **When Will AI Search Beat Google? 2025–2030 Forecast | TTMS**

<https://ttms.com/llm-powered-search-vs-traditional-search-2025-2030-forecast/>

⁸ ¹⁹ ⁴¹ ⁶⁹ **Goodbye Clicks, Hello AI: Zero-Click Search Redefines Marketing | Bain & Company**

<https://www.bain.com/insights/goodbye-clicks-hello-ai-zero-click-search-redefines-marketing/>

¹² ¹³ ²⁸ **'Amazon Rufus' AI experience comes to the Amazon Shopping app**

<https://www.aboutamazon.com/news/retail/amazon-rufus>

¹⁴ ¹⁵ ¹⁶ ²⁹ **Walmart: The Future of Shopping Is Agentic. Meet Sparky.**

<https://corporate.walmart.com/news/2025/06/06/walmart-the-future-of-shopping-is-agentic-meet-sparky>

¹⁷ ¹⁸ **Zillow debuts the only real estate app in ChatGPT - Oct 6, 2025**

<https://zillow.mediaroom.com/2025-10-06-Zillow-debuts-the-only-real-estate-app-in-ChatGPT>

²⁰ ²¹ ²² ²³ ²⁴ ²⁵ ⁸¹ ⁸⁶ ⁸⁷ **How to Optimize Your Website for AI Search: A Complete Guide to LLM-Friendly Web Design - The Chatbot Genius**

<https://thechatbotgenius.com/blog/llm-optimization-guide.html>

²⁶ ²⁷ **Build the web for agents, not agents for the web**

<https://arxiv.org/html/2506.10953v1>

³⁰ ³¹ **How AI and LLMs Are Ushering in a New Era of Conversational Commerce - Retail TouchPoints**

<https://www.retailtouchpoints.com/features/executive-viewpoints/how-ai-and-llms-are-ushering-in-a-new-era-of-conversational-commerce>

32 33 42 43 62 Before AI Exploits Our Chats, Let's Learn from Social Media Mistakes | TechPolicy.Press
<https://www.techpolicy.press/before-ai-exploits-our-chats-lets-learn-from-social-media-mistakes/>

34 35 36 37 38 Buy it in ChatGPT: Instant Checkout and the Agentic Commerce Protocol | OpenAI
<https://openai.com/index/buy-it-in-chatgpt/>

39 PayPal unveils partnership with OpenAI for payments on ChatGPT ...
<https://finance.yahoo.com/news/paypal-strikes-payments-wallet-deal-104230530.html>

40 You'll be able to pay with PayPal in ChatGPT next year - The Verge
<https://www.theverge.com/news/808079/chatgpt-paypal-openai-acp-purchases>

44 45 46 47 48 59 61 83 84 85 How Generative Engine Optimization (GEO) Rewrites the Rules of Search
| Andreessen Horowitz
<https://a16z.com/geo-over-seo/>

49 50 Introducing the GPT Store | OpenAI
<https://openai.com/index/introducing-the-gpt-store/>

51 The Ultimate Guide to Google Gemini Gems | by Leon Nicholls
<https://leonnicholls.medium.com/the-ultimate-guide-to-google-gemini-gems-78182be784af>

52 How To Use Gems in Gemini to Save Time with AI - YouTube
<https://www.youtube.com/watch?v=yO01B8OoXfo>

75 How AI Is Ending Google's Search Monopoly | by Gene Marks
<https://genemarks.medium.com/how-ai-is-ending-googles-search-monopoly-041554ec8027>